



Startup Profit Prediction (Best-Fit Model)

This app predicts startup profit and evaluates whether the model is best-fit.

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Investment.csv 2.4KB



Dataset Preview

	DigitalMarketing	Promotion	Research	State	Profit
0	165349.2	136897.8	471784.1	Hyderabad	192261.83
1	162597.7	151377.59	443898.53	Bangalore	191792.06
2	153441.51	101145.55	407934.54	Chennai	191050.39
3	144372.41	118671.85	383199.62	Hyderabad	182901.99
4	142107.34	91391.77	366168.42	Chennai	166187.94



Model Performance

Training R^2 (Bias): 0.9502

Testing R^2 (Variance): 0.9347



OLS Summary

OLS Regression Results

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Dep. Variable: Profit R-squared: 0.951

Model: OLS Adj. R-squared: 0.945
 Method: Least Squares F-statistic: 169.9
 Date: Thu, 18 Dec 2025 Prob (F-statistic): 1.34e-27
 Time: 18:09:20 Log-Likelihood: -525.38
 No. Observations: 50 AIC: 1063.
 Df Residuals: 44 BIC: 1074.
 Df Model: 5
 Covariance Type: nonrobust

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              coef  std err      t  P>|t|   [0.025   0.975]
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const      5.013e+04  6884.820   7.281   0.000   3.62e+04   6.4e+04
DigitalMarketing  0.8060   0.046  17.369   0.000   0.712   0.900
Promotion    -0.0270   0.052  -0.517   0.608  -0.132   0.078
Research     0.0270   0.017   1.574   0.123  -0.008   0.062
State_Chennai 198.7888 3371.007   0.059   0.953 -6595.030  6992.607
State_Hyderabad -41.8870 3256.039  -0.013   0.990 -6604.003  6520.229
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Omnibus:      14.782  Durbin-Watson:      1.283
Prob(Omnibus):    0.001  Jarque-Bera (JB):    21.266
Skew:           -0.948  Prob(JB):      2.41e-05
Kurtosis:        5.572  Cond. No.      1.45e+06
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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.45e+06. This might indicate that there are strong multicollinearity or other numerical problems.



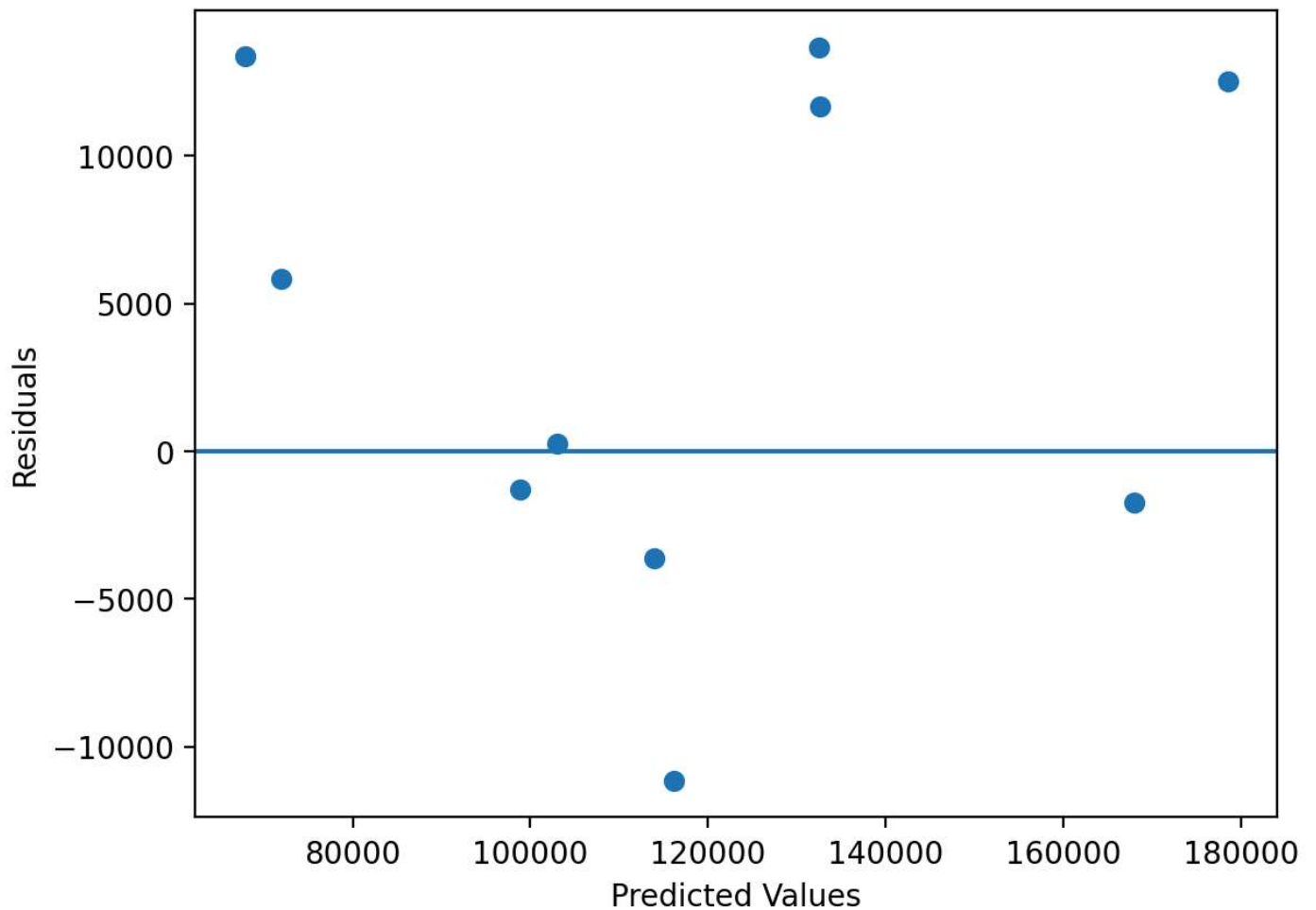
Best-Fit Evaluation

Adjusted R²: 0.9452

✓ Model is BEST FIT (Low Bias & Low Variance)



Residual Plot



Interpretation

- ✓ High Adjusted R^2 indicates strong explanatory power
- ✓ Small difference between training and testing scores shows good generalization
- ✓ Random residuals confirm no pattern